Predictive Analytics Report

Forecasting, predictive models, clustering, PCA, recommendations and limitations

Prepared by: Emmanuella Esi Arhin

Scope and data caveats

This report is one of four companion reports for the Data Science track (Data Quality, Exploratory Data Analysis, Statistical Analysis, Predictive Analytics), all produced by the executed notebook HealthConnect_DataScience_Notebook.ipynb. Five tables named in the brief were not supplied (Health Facilities master, Laboratory Tests, Prescriptions, Admissions, Referrals); only 5,000 of 60,000 bills, diagnoses and payments link to an appointment; and the data behave like simulated data. Analyses that depend on missing tables use labelled proxies or are marked not computable.

Predictive analytics

Patient volume and revenue forecasts (Tasks 1 and 2)

Each series has 12 monthly points, too few for seasonality. Four models (mean, naive, linear trend, exponential smoothing) were compared by rolling-origin back-testing; the plain mean is kept unless another model is at least 2% more accurate.

Series	Selected model	Trend p	2027 forecast total	Approx. 95% range
Monthly appointments (Appointments table)	Mean	0.489	5,000	4,682 - 5,318
Monthly appointments (management summary)	Mean	0.541	60,809	30,708 - 90,910
Monthly revenue (management summary, RWF)	Mean	0.789	1.13bn RWF	0.51bn RWF - 1.76bn RWF
Monthly billed revenue (linked bills, RWF)	Mean	0.370	0.77bn RWF	0.68bn RWF - 0.86bn RWF

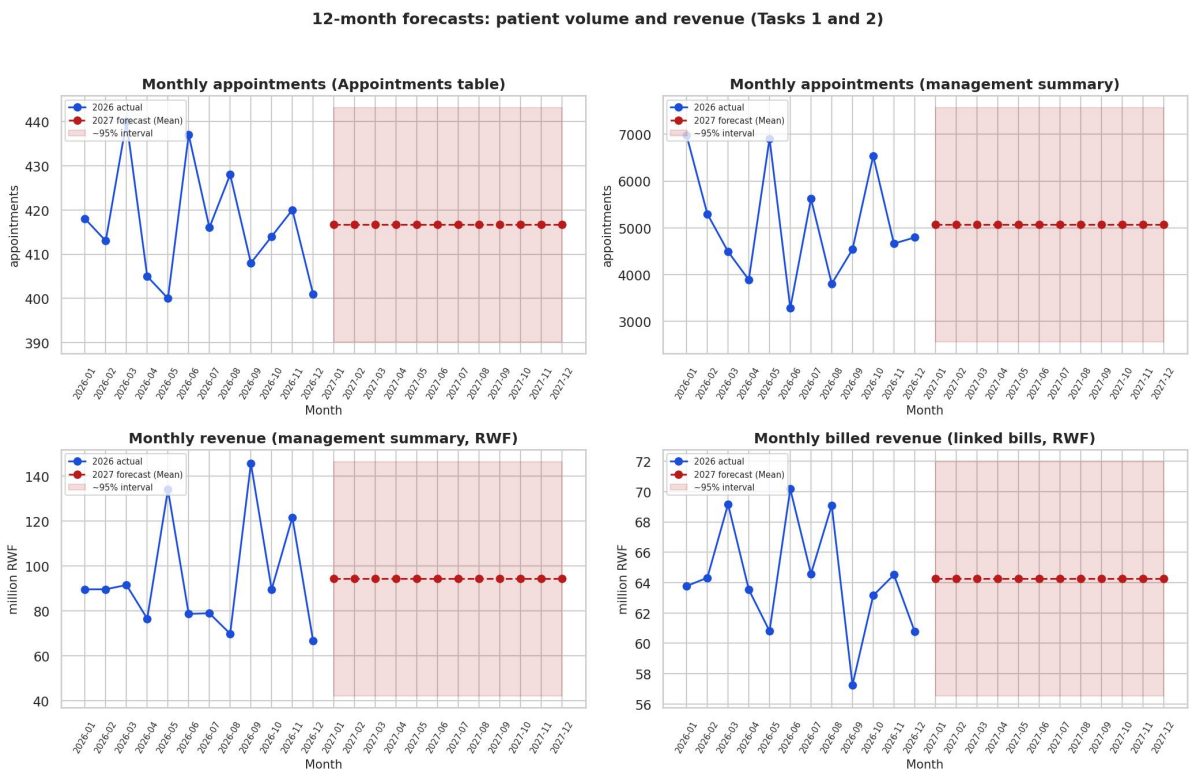


Figure: 12-month forecasts: patient volume and revenue (Tasks 1 and 2)

Back-testing found no model more than 2% more accurate than the plain historical mean for any of the four series, and none of the series has a significant trend (p = 0.37 to 0.79), so forecasts are flat at the 2026 average. The wide bands, especially for the management revenue summary (CV 27%), mean month-level revenue cannot be predicted reliably from 12 data points.

Department workload forecast (Task 5)

Department	Hist. monthly mean	2027 forecast	P(exceed busiest-10% month)
Cardiology	45.3	45.3	24%
Emergency	41.3	41.3	16%
Laboratory	41.6	41.6	13%
Pharmacy	40.8	40.8	10%
Pediatrics	40.9	40.9	9%
Radiology	43.8	43.8	8%
Internal Medicine	40.8	40.8	8%
Surgery	40.7	40.7	7%
Obstetrics & Gynecology	42.6	42.6	5%
ICU	38.9	38.9	0%

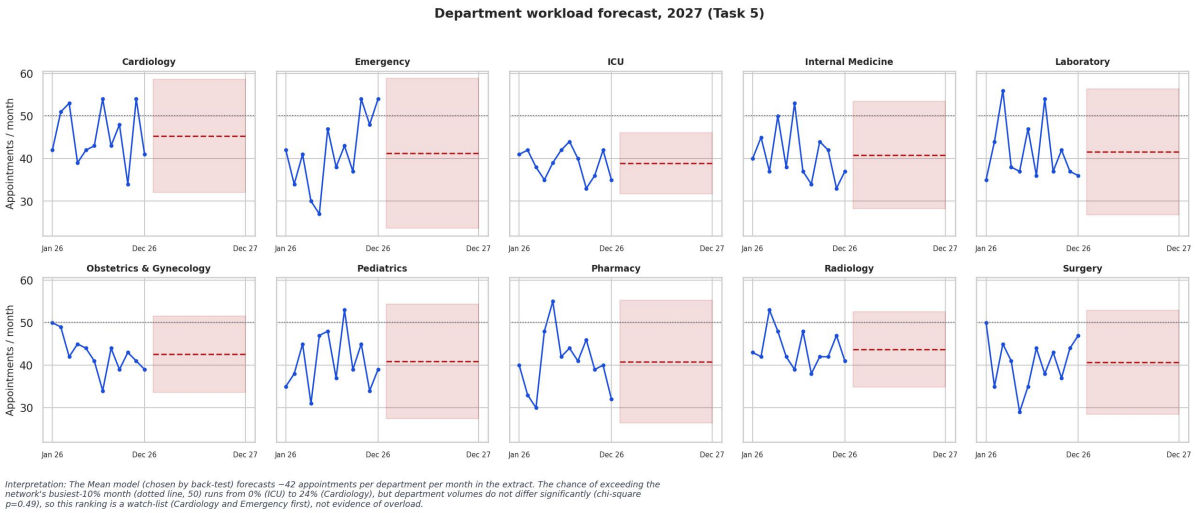


Figure: Department workload forecast, 2027 (Task 5)

The Mean model (chosen by back-test) forecasts ~42 appointments per department per month in the extract. The chance of exceeding the network's busiest-10% month (dotted line, 50) runs from 0% (ICU) to 24% (Cardiology), but department volumes do not differ significantly (chi-square p=0.49), so this ranking is a watch-list (Cardiology and Emergency first), not evidence of overload.

Admission prediction (Task 3, proxy target)

No Admissions table exists. The proxy target is an Emergency visit with a Severe diagnosis (10.9% of 5,000 appointments); visit type and severity were removed from the predictors. Validation: 5-fold patient-grouped cross-validation.

Model	AUC (95% CI)	PR-AUC	Precision	Recall	F1
Baseline (prevalence)	0.500	0.109	0.109	1.000	0.196
Logistic regression	0.494 (0.471-0.519)	0.112	0.112	0.884	0.199
Random forest	0.495 (0.470-0.522)	0.110	0.109	1.000	0.197
Gradient boosting	0.505 (0.479-0.531)	0.113	0.110	0.961	0.197

No model beats chance; an F1 of 0.20 is what flagging every visit achieves. Age, chronic status, diagnosis, department, facility and the other predictors carry no signal.

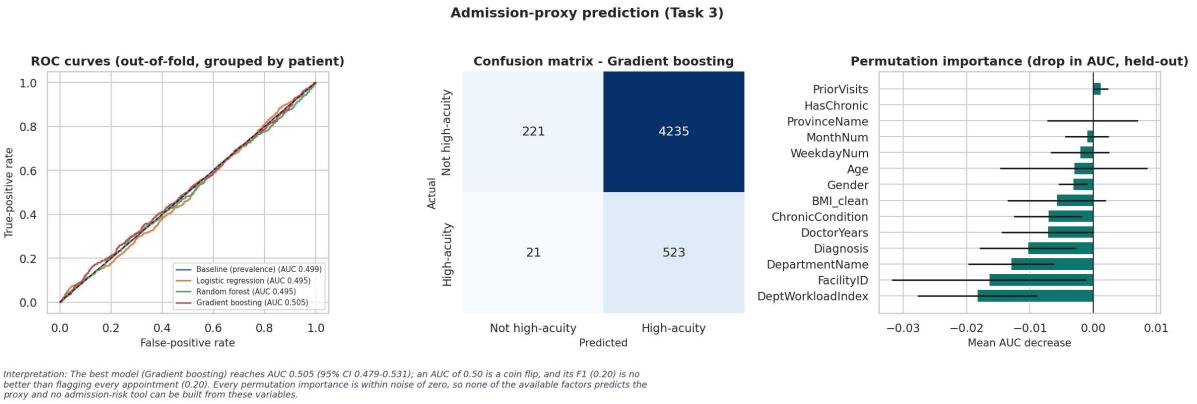


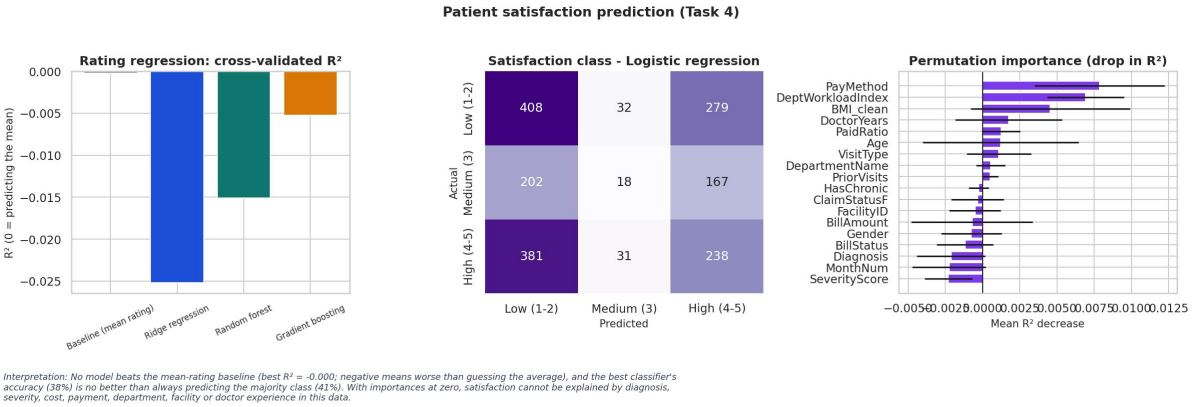
Figure: Admission-proxy prediction (Task 3)

The best model (Gradient boosting) reaches AUC 0.505 (95% CI 0.479-0.531); an AUC of 0.50 is a coin flip, and its F1 (0.20) is no better than flagging every appointment (0.20). Every permutation importance is within noise of zero, so none of the available factors predicts the proxy and no admission-risk tool can be built from these variables.

Patient satisfaction prediction (Task 4)

Regression model	R^2	MAE	RMSE
Baseline (mean rating)	-0.000	1.175	1.385
Ridge regression	-0.025	1.200	1.403
Random forest	-0.015	1.190	1.396
Gradient boosting	-0.005	1.183	1.389

Classifier (Low/Medium/High)	Accuracy	Balanced acc.	Macro-F1
Baseline (majority class)	0.409	0.333	0.194
Logistic regression	0.378	0.327	0.304
Random forest	0.409	0.343	0.290



Interpretation: No model beats the mean-rating baseline (best R² = -0.000; negative means worse than guessing the average), and the best classifier's accuracy (38%) is no better than always predicting the majority class (41%). With importances at zero, satisfaction cannot be explained by diagnosis, severity, cost, payment, department, facility or doctor experience in this data.

Figure: Patient satisfaction prediction (Task 4)

No model beats the mean-rating baseline (best R² = -0.000; negative means worse than guessing the average), and the best classifier's accuracy (38%) is no better than always predicting the majority class (41%). With importances at zero, satisfaction cannot be explained by diagnosis, severity, cost, payment, department, facility or doctor experience in this data.

Who is at risk of chronic disease?

A logistic regression on age, sex, adult BMI, province and blood group gives a cross-validated AUC of 0.516 (95% CI 0.507-0.526): statistically distinguishable from 0.5 but practically useless. Prevalence is about 75% in every adult BMI band (chi-square p = 1.00). No patient group is meaningfully more at risk in this data.

Clustering and PCA

Segmentation	k	Silhouette	Shuffled null	Stability (ARI)	Verdict
Patients (4,443)	4	0.22	0.18	0.55	Usable as visit-pattern segments; not natural clusters
Facilities (20)	2	0.21	0.17	0.92	Stable, modest: busier vs quieter sites
Departments (200)	6	0.16	0.17	0.38	No better than random; do not use

Patient segments

Segment (by profile)	Patients	Mean visits	Emergency share	Claim share	High-risk share
High Visits / High HasChronic	510	2.09	33%	70%	54%
Low EmergencyShare / High ClaimShare	1,873	1.00	0%	100%	39%

Segment (by profile)	Patients	Mean visits	Emergency share	Claim share	High-risk share
High EmergencyShare / High ClaimShare	902	1.00	100%	100%	40%
Low ClaimShare / Low Visits	1,158	1.00	34%	0%	36%

How management can use them: repeat visitors (about 510 patients, highest high-risk share) are candidates for care coordination and chronic follow-up; single-visit emergency patients for triage and primary-care referral; single-visit self-pay patients for insurance enrolment and financial counselling; routine insured visitors for preventive outreach. The high-risk share for repeat visitors is partly mechanical, since more visits give more chances of a severe diagnosis.

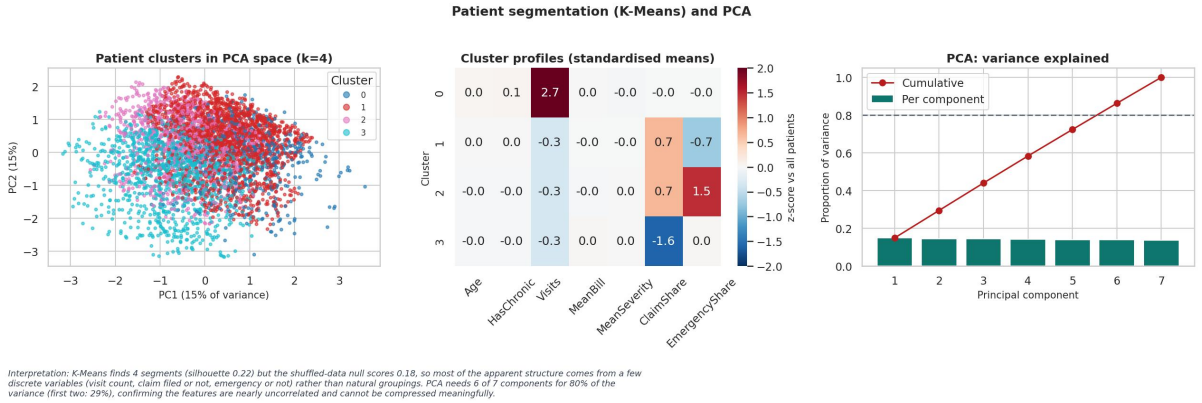


Figure: Patient segmentation (K-Means) and PCA

K-Means finds 4 segments (silhouette 0.22) but the shuffled-data null scores 0.18, so most of the apparent structure comes from a few discrete variables (visit count, claim filed or not, emergency or not) rather than natural groupings. PCA needs 6 of 7 components for 80% of the variance (first two: 29%), confirming the features are nearly uncorrelated and cannot be compressed meaningfully.

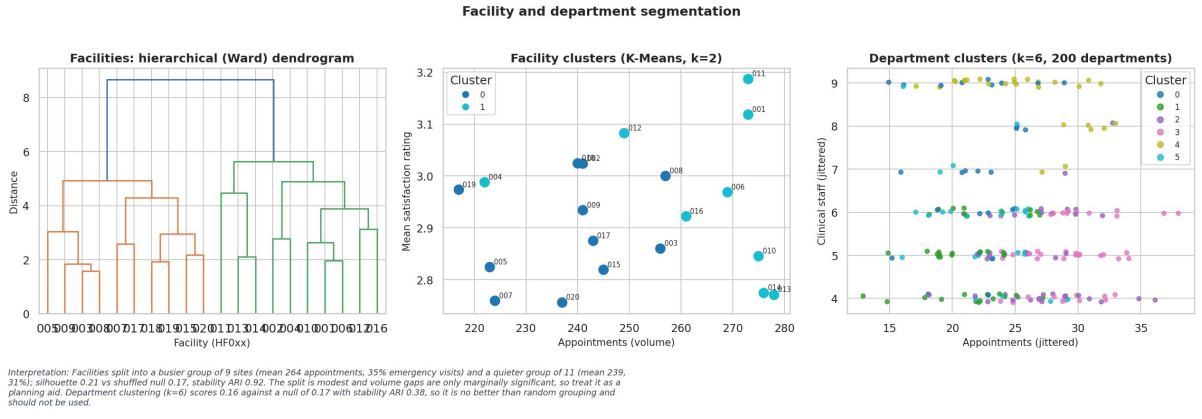


Figure: Facility and department segmentation

Facilities split into a busier group of 9 sites (mean 264 appointments, 35% emergency visits) and a quieter group of 11 (mean 239, 31%); silhouette 0.21 vs shuffled null 0.17, stability ARI 0.92.

The split is modest and volume gaps are only marginally significant, so treat it as a planning aid. Department clustering (k=6) scores 0.16 against a null of 0.17 with stability ARI 0.38, so it is no better than random grouping and should not be used.

PCA: 6 of 7 components are needed for 80% of the variance (first two: 29%), so the features cannot be compressed meaningfully.

Business recommendations

Area	Evidence	Recommendation	Confidence
Operational challenges	The transaction chain is broken: only 5,000 of 60,000 bills, diagnoses and payments link to an appointment, and facility/doctor/department disagree on ~95% of appointments.	Fix the data pipeline first: enforce appointment-ID foreign keys, derive facility from the department/doctor hierarchy, and add the missing admissions, lab, prescription, referral and facility-master tables before any further modelling.	High
Operational challenges	The brief's headline questions on admissions, length of stay, bed use and referrals cannot be answered: no admission data exist and all 2,000 beds are 'Available'.	Record bed status in real time (occupied/cleaning/out of service) and capture admission and discharge timestamps.	High
High-risk patient groups	75% of patients carry a chronic condition with no gradient by age, sex or province (tests 5-7, all non-significant after correction); 1,005 under-18s are recorded with hypertension. The rule-based flag marks 5,028 patients (25%) as high risk.	Audit chronic-condition coding (especially for children). Until validated, use a universal chronic-care outreach for flagged high-risk patients (age 65+ with a chronic condition, or any chronic condition with a severe diagnosis) rather than age/sex/region targeting.	Medium
Patient satisfaction	40% of all 21,064 ratings are 1-2. No facility or department differs significantly (tests 2 and 4), and no clinical, financial or operational variable predicts rating ($R^2 \leq 0$). Text comments contradict ratings in 30% of cases.	Treat poor satisfaction as a network-wide service problem, not a site problem: run a network-wide patient-journey review (waiting, communication, billing). Link every survey to its appointment, and replace free-text comments with structured reason codes.	Medium

Area	Evidence	Recommendation	Confidence
Resource optimisation	Doctor workload ranges 5-28 appointments (P90/P10 = 1.9x); facility load per clinical staff member differs by 32% (HF013 highest, HF004 lowest), while rosters are almost identical (57-60 staff). Facility volume gaps are marginal (p=0.052).	Pilot a scheduling rule that balances doctor panels, and move 2-3 clinical staff from the lowest-load sites (HF004, HF005, HF007) to the highest-load sites (HF013, HF014, HF011); re-measure after one quarter before a permanent change.	Low-Medium
Revenue improvement	0.92bn RWF (9.9% of billings) is unpaid across 11,887 underpaid bills, yet Billing.Status is unrelated to what was paid (test 9). 21,028 insurance claims (2.44bn RWF) are still pending, spread evenly over four insurers.	Rebuild payment status from payment records (Amount - AmountPaid), start a collections list for the largest underpaid bills, and set a claim turnaround target for all four insurers (e.g. clear pending claims within 30 days).	High
Revenue improvement	The management revenue summary (1.13bn RWF) does not reconcile with billing (9.31bn RWF) and swings 66M-146M a month independent of volume (rho = 0.33, p = 0.30).	Reconcile finance reporting to the billing ledger monthly and explain the May, September and November spikes before using revenue forecasts for budgeting.	High
Expanding services	Demand is flat (no trend in monthly appointments, diagnoses or revenue; all p > 0.05) and no department, diagnosis or facility is growing. Cardiology (544) and Radiology (525) are the busiest department types but not significantly so.	Do not commit capital to expansion on this evidence. Re-run the demand forecast once 24+ months of complete data exist, and use Cardiology and Radiology as the first capacity-review candidates.	Medium

Limitations

- Missing tables block outcome analyses; proxies are labelled and should not be used clinically.
- Only 8.3% of billing, diagnosis and payment records link to an appointment, so appointment-level findings rest on 5,000 rows.
- Twelve months cannot support seasonality; forecasts are flat baselines with wide intervals.

- Future-dated records: 874 appointments fall after the analysis date (24 Oct 2026) yet carry diagnoses and bills. They were retained and flagged.
- The data look simulated; real-world conclusions should wait until source systems are validated.